

Cryptography and Network Security

CRT

Today's Lecture

- **Chinese Remainder Theorem (CRT)**

- The Chinese Remainder Theorem is used to solve a **set of congruence equations** with **one variable**.
- Each equation has a **different modulus**, and all moduli are **relatively prime** to each other.
- It helps to **break a large problem** with a big modulus into **smaller and simpler problems**.
- After solving the smaller problems, CRT **combines the results** to get the final answer.

- **Applications:**

- Cryptography
- Fast mathematical computations
- Error-correcting codes

- CRT provides an **efficient method to reconstruct a number** when its remainders with respect to different moduli are known.

The Problem

- Jay has a basket full of apples. When he takes the apples out of the basket 2 at a time, there is 1 apple left over. When he takes them out 3 at a time, there are 2 apples left over. Likewise, when he takes the apples out 4, 5, and 6 at a time, he finds remainders of 3, 4, and 5, respectively. However, when he takes the apples out 7 at a time, there are no apples left over.
 - What is the least amount of apples that could be in Jay's basket?

Chinese Remainder Theorem

If you have integers $m_1, m_2, \dots, m_k$ that are pairwise coprime, and given remainders $a_1, a_2, \dots, a_k$ for each,

Then the system of congruences

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ x \equiv a_2 \pmod{m_2} \\ \vdots \\ x \equiv a_k \pmod{m_k} \end{cases}$$

has exactly one solution modulo $M = m_1 m_2 \dots m_k$.

The solution is:

$$x \equiv \sum_{i=1}^k a_i M_i y_i \pmod{M} \quad \text{Here } (y_i \equiv M_i^{-1} \pmod{m_i})$$

where

$$M_i = M/m_i$$

and y_i is the modular inverse of M_i modulo m_i , i.e. $M_i y_i \equiv 1 \pmod{m_i}$.

Example: Solve the following simultaneous linear congruences?

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

Step 1: Set up the CRT notation

Let:

$$m_1 = 3, a_1 = 2; m_2 = 5, a_2 = 3; m_3 = 7, a_3 = 2$$

Moduli are pairwise coprime: .

Step 2: Compute the total modulus

$$M = m_1 \times m_2 \times m_3 = 3 \times 5 \times 7 = 105$$

Step 3: Compute M_i

$$M_1 = \frac{M}{m_1} = \frac{105}{3} = 35$$

$$M_2 = \frac{M}{m_2} = \frac{105}{5} = 21$$

$$M_3 = \frac{M}{m_3} = \frac{105}{7} = 15$$

Step 4: Compute inverses $y_i \equiv M_i^{-1} \pmod{m_i}$

We solve:

$$M_i \cdot y_i \equiv 1 \pmod{m_i}$$

For $M_1 = 35$ in mod 3:

$$35 \pmod{3} = 2, 2y_1 \equiv 1 \pmod{3}$$

$$\text{Trial: } 2 \times 2 = 4 \equiv 1 \pmod{3} \Rightarrow y_1 = 2.$$

For $M_2 = 21$ in mod 5:

$$21 \pmod{5} = 1, 1 \cdot y_2 \equiv 1 \pmod{5} \Rightarrow y_2 = 1.$$

For $M_3 = 15$ in mod 7:

$$15 \pmod{7} = 1, 1 \cdot y_3 \equiv 1 \pmod{7} \Rightarrow y_3 = 1.$$

Step 5: Apply CRT formula

The unique solution modulo M is:

$$x \equiv (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \pmod{M}$$

Substitute:

$$x \equiv (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1) \pmod{105}$$

$$x \equiv 140 + 63 + 30 \pmod{105}$$

$$x \equiv 233 \pmod{105}$$

Step 6: Reduce to smallest non-negative solution

$$233 \div 105 \text{ gives remainder } 233 - 2 \times 105 = 23$$

Thus:

$$x \equiv 23 \pmod{105}$$

Verification

$$23 \div 3 = 7 \text{ remainder } 2 \checkmark$$

$$23 \div 5 = 4 \text{ remainder } 3 \checkmark$$

$$23 \div 7 = 3 \text{ remainder } 2 \checkmark$$

Final Answer

$$\boxed{x \equiv 23 \pmod{105}}$$

Exercise

Problem 1

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{4} \end{cases}$$

Problem 2

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{5} \\ x \equiv 1 \pmod{2} \end{cases}$$

Problem 3

$$\begin{cases} x \equiv 10 \pmod{17} \\ x \equiv 4 \pmod{19} \\ x \equiv 1 \pmod{23} \end{cases}$$

Problem 4

Find an integer that has a remainder of 3 when divided by 2 and 5, but is divisible by 7.

Example

Assume we need to calculate $z = x + y$ where $x = 123$ and $y = 334$, but our system accepts only numbers less than 100. These numbers can be represented as follows:

$$\begin{array}{ll} x \equiv 24 \pmod{99} & y \equiv 37 \pmod{99} \\ x \equiv 25 \pmod{98} & y \equiv 40 \pmod{98} \\ x \equiv 26 \pmod{97} & y \equiv 43 \pmod{97} \end{array}$$

Adding each congruence in x with the corresponding congruence in y gives

$$\begin{array}{ll} x + y \equiv 61 \pmod{99} & \rightarrow z \equiv 61 \pmod{99} \\ x + y \equiv 65 \pmod{98} & \rightarrow z \equiv 65 \pmod{98} \\ x + y \equiv 69 \pmod{97} & \rightarrow z \equiv 69 \pmod{97} \end{array}$$

Now three equations can be solved using the Chinese remainder theorem to find z . One of the acceptable answers is $z = 457$.